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ML Project from Roger

2 messages

Jyothi Shetty <jyothis@rvce.edu.in> Wed, Nov 26, 2025 at 9:15 AM
To: SKANDA P R <skandapr.cs22@rvce.edu.in>
Cc: SUHAS RAJ H R <suhasrajhr.cs22@rvce.edu.in>, G Shobha <gshobha@rvu.edu.in>

Hi Skanda,

Thank you for agreeing to work on the project. The problem statement is here. You can form a team of 4 students and start NOW in 7th sem if feasible. Let me know your thoughts about it accordingly I can schedule a meeting with Roger.

Project 4 – DPR: Human-Supervised Multi-Agent Collaboration Protocol

Working Title:
"Designing a Scalable Protocol for Human-Supervised Collaborative Problem Solving by Multiple AI Agents"

Note: We will provide full design specifications for both phases.

Core Goal

Implement the **Distributed Protocol for Reasoning (DPR)** — a structured communication and governance system that enables a human supervisor to coordinate **2–10 AI agents** working together on shared problem-solving tasks.
The project focuses on two configurations:

- **Phase 1:** 2–4 agents (small-team deep reasoning)
- **Phase 2:** 5–10 agents (medium-team distributed reasoning with governance)

Students will build the DPR apparatus, implement turn-taking, hand-raising, and facilitator logic, and evaluate the stability, fairness, and effectiveness of multi-agent collaboration.

Research Questions

1. **Structured Coordination:**
   Does enforcing turn-taking (talking-stick protocol) improve clarity and reduce conversational interference among multiple AIs?
2. **Human Supervision:**
   How effectively can a human guide multi-agent reasoning using pause, redirect, and inject commands?
3. **Scalability & Stability:**
   How does collaborative coherence change as the number of AI agents increases from 2 to 10?

4. **Governance Mechanisms:**
   Do hand-raise interrupts, quotas, and facilitators prevent domination, loops, or agent starvation?

Scope / Tasks

Phase 1 (2–4 agents)

Implement the small-team core protocol:

- Talking-stick round-robin turn-taking
- Logged “ignored responses” for transparency
- Human console: **PAUSE / RESUME / REDIRECT / INJECT**
- Basic fairness and sequencing

Phase 2 (5–10 agents)

Scale the protocol to medium-sized groups using governance tools:

- Hand-raise interrupt mechanism
- Quota-based contribution economy
- Priority ordering (proximity to token)
- Anti-starvation rules
- **Rule-based Facilitator** (Phase 3.5): loop detection, fairness, redirection
- (Optional) **AI-assisted Facilitator** (Phase 4)

Evaluation

Students perform controlled experiments comparing:

- Structured DPR vs. unstructured multi-agent chat
- Small-team (Phase 1) vs. medium-team (Phase 2) behavior
- Impact of facilitator interventions on stability and progress
- Fairness, turn distribution, and response quality metrics

Deliverables

- **DPR Implementation** (Python): message bus, token controller, interrupt queue, facilitator module
- **Supervisor Console** for human-in-the-loop guidance
- **Experiment Suite** for 2–10 agent configurations
- **Research Report (5–8 pages)** covering architecture, experiments, and findings

- **Demonstration**: multi-agent problem-solving session using DPR

Why It's a Clean, Independent Project

DPR is **completely independent** of the MFN sequence and can be completed immediately. It is engineering-heavy, supervision-oriented, and produces publishable results in:

- multi-agent systems,
- AI governance,
- coordinated reasoning, and
- human-AI interaction.

Thanks & Regards

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SKANDA P R <skandapr.cs22@rvce.edu.in> Fri, Nov 28, 2025 at 9:06 PM
To: SUHAS GOWDA L <suhasgowdal.cs22@rvce.edu.in>, PUNEETH B <puneethb.cd22@rvce.edu.in>

Hello Guys, this is the project information, please keep it confidential, at least until 8th Sem.
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